

GROUP DETAILS

IN AN EMERGENCY CALL:

Leader:		Group Phone:	
Participants:			

ACTIVITY DETAILS

Purpose of Activity:		Date:	
Benefit of Activity: Brings scientific concepts to life through hands on investigation. Builds precise practical skills and respect for lab protocols. Develops critical thinking, hypothesis testing, and data literacy. Encourages curiosity and creativity in problem solving. Links learning to real world STEM pathways and careers. Strengthens teamwork and communication during shared experiments.			

HAZARDS

HAZARD	RISK	RISK BENEFIT	MEASURE	RISK TO	RISK LEVEL
Chemical handling, contact, or ingestion	Chemical burns, poisoning, skin irritation, accidental ingestion	Handling chemicals under supervision builds respect for substances, teaches industry-level safety, and makes science engaging and realistic.	Store chemicals securely with clear labelling. Train pupils in handling, use PPE, never taste chemicals, and wash hands after experiments. Provide eyewash stations and gloves. (ALL)	Students, staff	Before Measure: High After Measure: Med
Explosion or high-energy reactions	Burns, hearing damage, flying debris	Shows chemistry's power safely, inspiring curiosity and respect for controlled experimentation.	Use safety screens, shields, and small quantities. Demonstrate risky experiments first and maintain strict supervision. (ALL)	Students, staff	Before Measure: High After Measure: Med

HAZARD	RISK	RISK BENEFIT	MEASURE	RISK TO	RISK LEVEL
Fire hazards	Burns, smoke inhalation, property damage	Students learn that risk is manageable, not avoided, building practical safety skills transferable to future study or work.	Keep extinguishers, blankets, and isolation switches accessible. Remove combustibles from benches, store flammables safely, and rehearse fire evacuation drills. (ALL)	Students, staff	Before Measure: High After Measure: Med
Fumes, ventilation failure, and airborne hazards	Respiratory irritation, poisoning	Using extraction equipment shows pupils how scientists work safely with hazardous materials in controlled environments.	Use fume cupboards for volatile substances, ensure ventilation is functional, and teach proper use. Have backup extraction protocols and emergency evacuation plans. (ALL)	Students, staff	Before Measure: High After Measure: Med
Gas supply and leaks	Fire, explosion, poisoning	Teaches pupils how to safely manage gas systems, preparing them for advanced scientific study.	Fit isolation valves, train staff in odour recognition and evacuation, lock taps when not in use, and check pipes regularly. (ALL)	Students, staff	Before Measure: High After Measure: Med
Group behaviour and management	Disruption, unsafe behaviour, injury	Large group experiments teach teamwork, communication, and responsibility in a realistic lab setting, preparing students for collaborative science work.	Set clear lab rules, seating plans, and maximum group sizes based on activity risk. Assign roles, supervise closely, and stop work immediately if safety rules are broken. (ALL)	Students, staff	Before Measure: High After Measure: Med
Heat sources and burns	Burns, fire risk	Exposure to heat builds confidence and skill in handling equipment safely, a core part of scientific training.	Teach safe lighting of Bunsen burners, use heatproof mats, supervise ignition closely, and ensure appropriate clothing. Keep first-aid and burn kits available. (ALL)	Students, staff	Before Measure: High After Measure: Med

HAZARD	RISK	RISK BENEFIT	MEASURE	RISK TO	RISK LEVEL
Spillages and chemical reactions	Slips, burns, toxic fumes	Controlled exposure to accidents teaches students emergency responses, building confidence and maturity around hazards.	Provide spill kits, neutralisers, and clear procedures. Demonstrate riskier reactions before students attempt them. Keep quantities small and shield high-risk setups. (ALL)	Students, staff	Before Measure: High After Measure: Med
Animal handling	Bites, scratches, allergic reactions	Builds empathy, responsibility, and understanding of life sciences through safe, hands-on experiences.	Supervise closely, follow welfare guidance, limit handling, and enforce handwashing after contact. Quarantine animals as needed. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Biological specimens and allergens	Allergic reactions, infection	Encourages empathy for living organisms and teaches biological safety and allergy awareness, boosting STEM engagement.	Follow hygiene rules, use gloves, wash hands, and disinfect workspaces. Screen for allergies, quarantine samples, and dispose of waste safely. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Electrical hazards and power outages	Electric shock, equipment damage, experiment failure	Teaches safe handling of electrical devices and resilience when equipment malfunctions, preparing students for real-world labs.	PAT test all equipment, use RCDs, keep wires dry, and unplug unused devices. Train pupils to safely stop experiments if power is lost unexpectedly. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Emergency preparedness	Delayed response, increased injury risk	Gives pupils transferable life skills and confidence to manage risk in unfamiliar environments.	Display safety signage, train students on exits, alarms, and first-aid points, and run annual drills for spills, fires, and chemical exposure. (ALL)	Students, staff	Before Measure: Med After Measure: Low

HAZARD	RISK	RISK BENEFIT	MEASURE	RISK TO	RISK LEVEL
Glassware breakage and sharps	Cuts, puncture wounds, infection	Real glassware teaches careful handling, precision, and responsibility, preparing students for advanced science environments.	Use safety glass or plastic where possible, provide glass disposal bins, sweep kits, and supervise closely. Train pupils to carry glassware upright and store equipment safely. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Inexperienced staff or insufficient training	Incorrect supervision, accidents	Allowing staff to lead experiments builds their confidence and skill, helping schools run engaging lessons without relying solely on external specialists.	Provide regular lab safety training, induction for new staff, mentoring for inexperienced teachers, and ensure only trained staff supervise high-risk experiments. Maintain easy access to safety manuals. (ALL)	Staff	Before Measure: Med After Measure: Low
Lasers, UV, and light experiments	Eye damage, burns	Practical optics and light experiments spark interest in physics and show how precision tools are safely applied.	Use low-power lasers, shields, and warning signs. Enforce eye protection and beam stops. Store sources securely and supervise all use. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Noise and impact from equipment	Hearing damage, injuries	Exposure to realistic lab environments builds resilience and mirrors industry conditions, keeping science exciting.	Warn students in advance of loud reactions, use hearing protection where needed, and supervise mechanical equipment. (ALL)	Students, staff	Before Measure: Med After Measure: Low
PPE misuse or fatigue	Reduced protection, complacency	Teaches discipline and focus, reinforcing that PPE supports—not replaces—safe behaviour.	Train students in correct PPE use, replace damaged gear, and regularly remind them to stay alert despite PPE comfort. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Sharp instruments and cutting tools	Cuts, puncture wounds, infection	Handling scalpels and cutters builds dexterity, patience, and awareness of risks, essential for STEM pathways.	Provide age-appropriate tools, teach cutting techniques, supervise use, and dispose of sharps in designated bins. (ALL)	Students, staff	Before Measure: Med After Measure: Low

HAZARD	RISK	RISK BENEFIT	MEASURE	RISK TO	RISK LEVEL
Structural hazards and storage	Falling equipment, injuries	Teaches students about lab infrastructure and the importance of safe facility maintenance.	Inspect shelving, ceiling fittings, and lab furniture quarterly. Use secured storage for heavy equipment and chemicals. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Trip hazards and clutter	Trips, falls, injuries	Builds organisational skills and situational awareness, promoting professionalism in science.	Keep aisles clear, store bags away, and inspect benches and stools. Enforce a tidy-as-you-go policy and address hazards immediately. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Waste disposal and sustainability	Pollution, chemical hazards	Teaches environmental responsibility and the real world impact of safe lab practices.	Segregate waste into labelled bins, autoclave biological samples, and follow chemical disposal regulations. Educate pupils on environmental impacts. (ALL)	Students, staff	Before Measure: Med After Measure: Low
Unpredicted risks	Illness, injury, death		Ongoing dynamic risk assessment carried out by all staff. Any observed potential hazards should be notified to group leader and immediate action taken if required. (ALL)	All	N/A

NOTES

Extra notes & activity evaluation:

Completed by

Reviewed/Approved by

Risk Assessment Date

Review Required Date